

Table of contents

Introduction

TOPIC 1: The Computer

TOPIC 2: Types of Computers

TOPIC 3: Fundamental Elements of a Computer

Introduction



We all know that computers are essential pieces of technology that we use to make life easier and perform jobs faster.

A computer increases our productivity, connects us to the internet, lets us talk with people across the world, helps us store a lot of information, and even make money!

In this lesson, we will start with the **basics**.

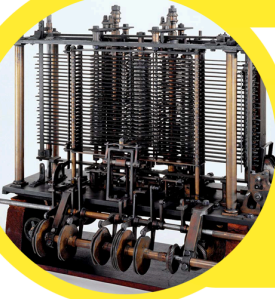
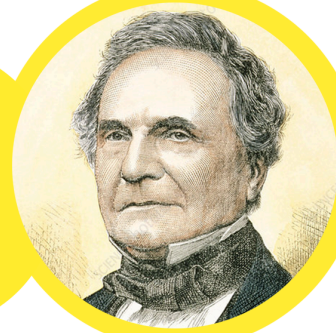
We want to understand how the computer works, how many types of computers are there, what are the functions of the computer, and more!

TOPIC 1: The Computer

In technical terms, the **computer** is a device that accepts information in the form of digitalized data, and manipulates it for some result based on a program or sequence of instructions on how the data is to be processed.

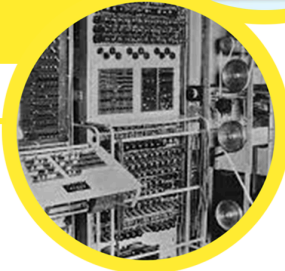
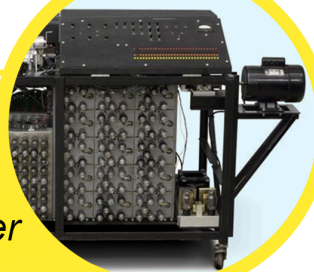
A BRIEF HISTORY OF THE COMPUTER

The computer we know today is a result of various improvements made since the 19th century by an English mathematics professor named **Charles Babbage**.



Babbage designed the **Analytical Engine** where the basic framework of computers today are based on.

The first electronic digital computer in 1937 was the **Atanasoff-Berry Computer** or ABC. In 1943, a computer called **Colossus** was built for the military.



Developments continued until 1946 when the first general-purpose digital computer was made. Over the course of 1947 to 1962, the computer was introduced to the **public** and **programming languages** were then developed. Computers had memory, operating systems, tapes, disks, and printers.



Since 1963, the computers started becoming smaller and more powerful. The **PC** or personal computer we know today is introduced in 1980 from the **MS-Dos**.



*Improvements were slowly made to give us the **Windows** operating system we know today and the **Macintosh** computer from Apple.*



The computer is still being improved today and many functions are still being developed to make human tasks easier and faster.

courtesy: britannica, computerhistory.org, hp computer museum

TOPIC 2: Types of Computers

1. Supercomputer

Most powerful and most advance in terms of performance and data processing. These computers are commonly used by large organizations to conduct specific tasks such as *weather forecasting, research, and exploration*.

Ex. IBM's Mira in United States. IBM's Sequoia in United States



Fujitsu K Computer in Japan

2. Mainframe Computer

Large and expensive computer capable of handling thousands of users and programs simultaneously. Mainframe computers can handle, process, and store large amount of data. These are commonly stored in large air-conditioned rooms because of its big size.

Ex. Fujitsu's ICL VME, Hitachi's Z800



Fujitsu ICL VME

3. Minicomputer

Also known as midrange computers. It features all the functions of a large size computer, but has a smaller size. Minicomputers can support many users because of its multi-processing system.

Ex. K-202, IBM Midrange computers, SDS-92



K-202

4. Microcomputer

Most widely used and fastest growing type of computer. Desktop computers, laptops, tablets, and smartphones are types of microcomputers.

TOPIC 3: Fundamental Elements of a Computer

THE 3 FUNDAMENTAL ELEMENTS OF THE COMPUTER

- **System Unit**>acts like the center or core, processing the data and information it receives from input devices.
- **Input Devices**>An **input device** is any hardware **device** that sends data to a computer, allowing you to interact with and control the computer.
- **Output Devices**>these are the devices like printers. It receives the system unit's processed information.

